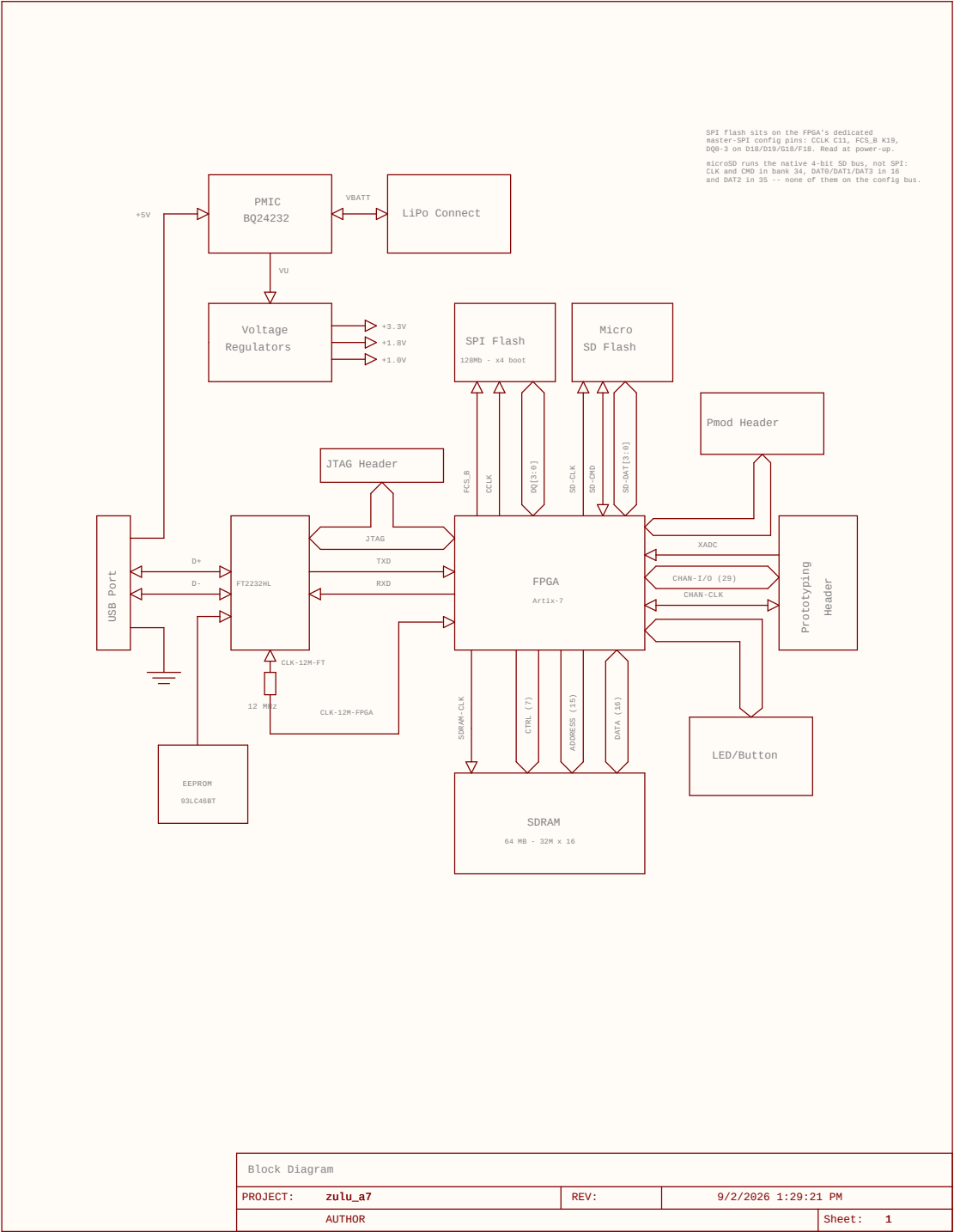
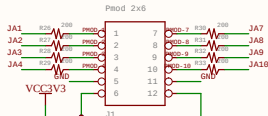
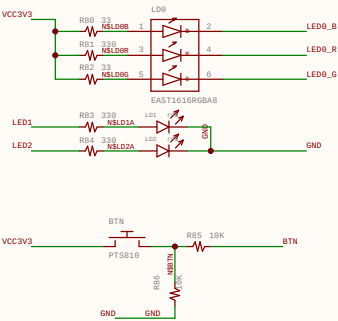
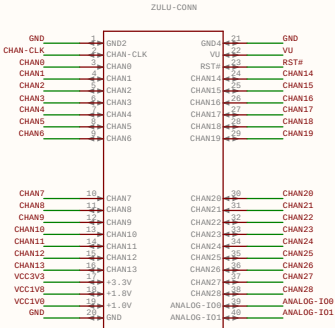




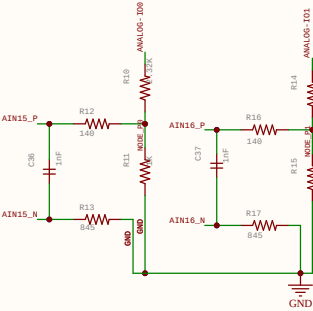


Power Supplies			
PROJECT:	zulu_a7	REV:	9/2/2026 1:29:21 PM
AUTHOR			Sheet: 2

X2: 40 pins on a 2.54 mm grid, rows 22.86 mm (0.900 in) apart, numbered 1-40 straight through from the right and skipping both landings. The two rows are mirror images. TOP ROW, x 50.69 down to 1.27: GND (pin 1), CHAN-CLK (2), CHAN0-CHAN6 (3-9), then CHAN7-CHAN13 (10-16), +3.3V, +1.8V, +1.0V, GND (17-20), then CHAN14-CHAN16 (21-23), CHAN17-CHAN19 (24-26), CHAN20-CHAN28 (29-36), ANALOG-IO0, ANALOG-IO1 (37, 38), CHAN17-CHAN19 (27-29), then CHAN20-CHAN28 (30-38), ANALOG-IO0, ANALOG-IO1 (39, 40). FOUR positions in each row -- x 36.83, 34.29, 31.75, 29.21, between pins 9 and 10 above and between pins 29 and 30 below -- are RESERVED, not spare: the edge-mounted micro-USB X1 lands on the top row (7.88 mm of copper), the top-side L1Po header X4 (JST B2B-PH-SMA-1B, 7.95 mm body) lands on the bottom row directly opposite it.



Digilent 12-pin Pmod (JA). Pins 1-4 and 7-10 are signals through 200R series resistors; pins 5/11 are GND and 6/12 are VCC3V3 -- so a 2x6 header carries EIGHT signals, not twelve, which is why the names skip JA5/JA6. No decoupling here: the Pmod spec puts bypassing on the module, and VCC3V3 already carries ~170uF. The Cmd A7 has none at its Pmod either. JA1-JA4 (the top row, the half A Pmod uses for SPI) are all in bank 16: A14 A15 B15 A16, A17 is C15, also bank 16. JA8/JA9/JA10 are W16/W2/W7 -- banks 14/35/34 -- because nothing else was free. All five banks are VCC0=3.3V, so this costs timing coherence, not levels.



© 2026, Christian Tamba. Some rights reserved.

CC BY SA Licensed under a Creative Commons Attribution-Share-Alike 4.0 International License. See creativecommons.org/licenses/by-sa/4.0/

General IO

PROJECT:	zulu_a7	REV:	9/2/2026 1:29:21 PM
AUTHOR	Sheet: 3		

U3 = AS4C32M16SB-7TCN: 512Mbit (32M x 16) = 64MB.
Pin-for-pin and dimensionally identical to the 256MB
AS4C16M16SA-6TIN it replaces -- all 54 pins and every
TSP: 1 dimension checked against both datasheets, and
pin 40 is NC on both.
One DIFFERENCE, and IT IS NOT A PIN: the column address
is A0-A9 (1024 columns), not A0-A8 (512). The memory
controller must be set to 10 column bits, or it reaches
only half the array. Row A0-A12, banks BA0/BA1 and
refresh (8192 cycles / 64ms) are unchanged.
-7TCN since 2026-09-08 (-6TIN dry): 143 MHz max at CL13,
tRCD/tRP 21, tRC 63 ns, TA 0-70 C, ID01 max 110 mA.

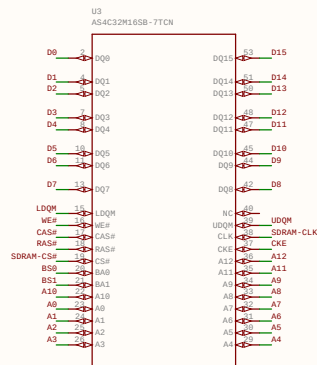
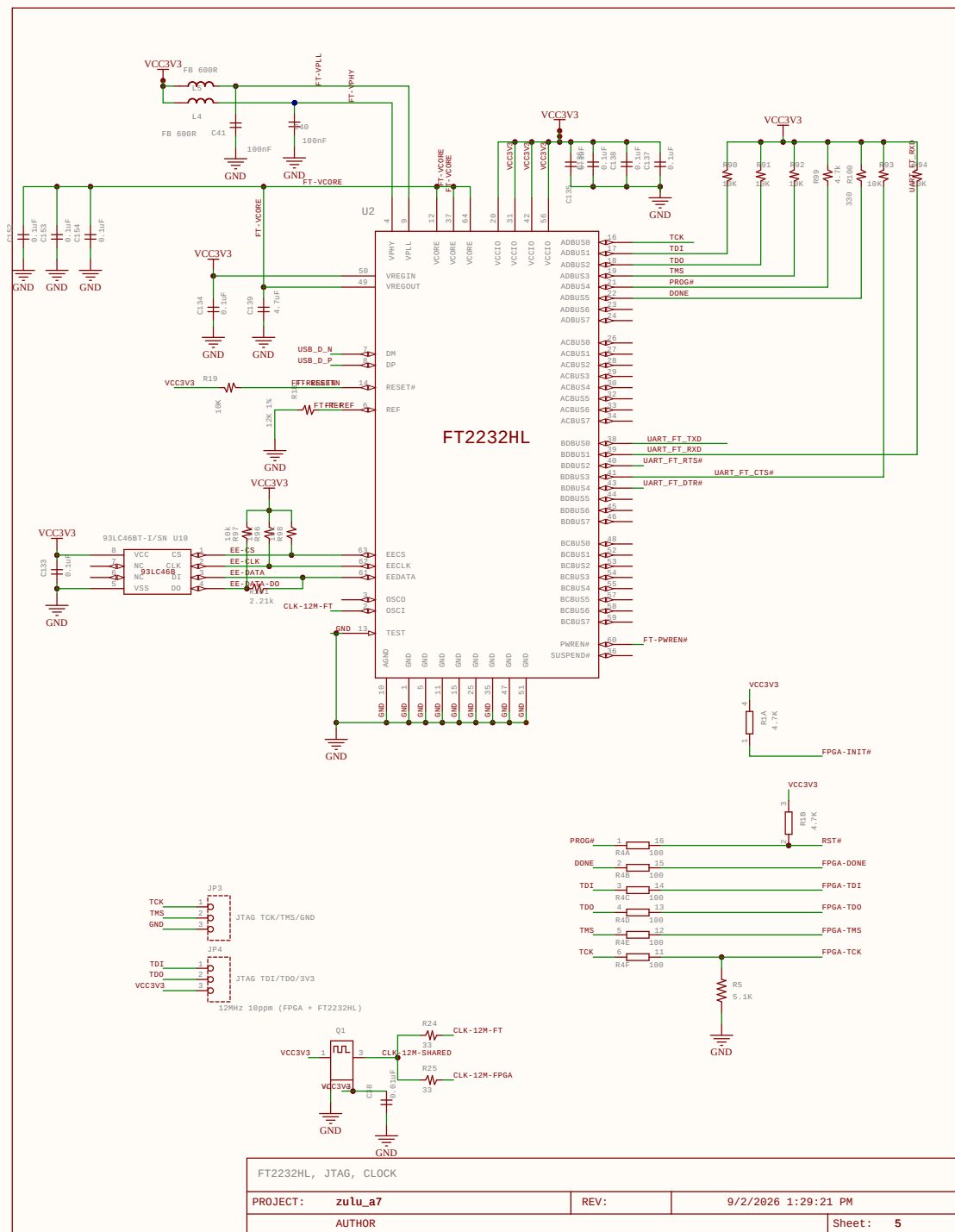
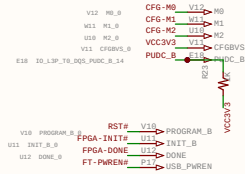


Figure 1: Schematic representation of the VSSQs. The figure shows eight horizontal lines representing VSSQs, each with a label on the left and a corresponding VCC3V3 line on the right. The labels are VSSQ#3, VSSQ#1, VSSQ#4, VSSQ#2, VSSQ#5, VSSQ#2, VSSQ#1, and VSSQ#3. The VCC3V3 lines are labeled with their respective pin numbers: 1, 3, 9, 14, 27, 43, and 49. Each VSSQ line is connected to a GND pin on the left and a VCC3V3 pin on the right. The VSSQ#2 and VSSQ#1 lines are also connected to a C4 pin on the left and a C5 pin on the right. The VSSQ#5 line is connected to a C6 pin on the left and a C7 pin on the right. The VSSQ#2 line is connected to a C8 pin on the left and a C9 pin on the right. The VSSQ#3 line is connected to a C10 pin on the left and a C11 pin on the right.

Memory			
PROJECT:	zulu_a7	REV:	9/2/2026 1:29:21 PM
AUTHOR			Sheet: 4



CONFIG



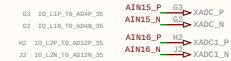
CLK



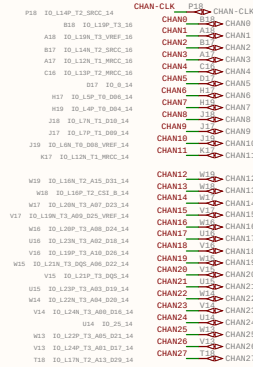
JTAG



ANALOG



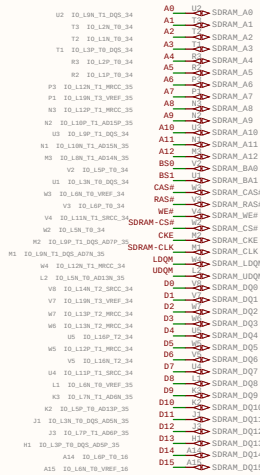
PROTOTYPING HEADER



The six pins above CLK_FLASH are the ONLY balls master-SPI configuration can use (UG479). They are ordinary user I/O once DONE goes high. SDRAM A4 moved K10 -> D17 to Free FCS_B; the ESP reset moved D17 -> B15. microSD is on its own CLK/CMD/DAT0 so a card cannot fight the configuration bus.

The six microSD signals below CLK_FLASH are the native 4-bit SD bus: CLK CMD DAT0-DAT3. All sit in bank 34 byte group T0 (R2 T1 T2 U1 V2 W2). DAT1/DAT2 took CHAN1/CHAN0 off the prototyping header; DAT3 is the old SPI chip-select renamed, so only two pins were spent.

SDRAM



UNCONNECTED BALLS
XADC analog (unused)

Thermal diode (unused)

Battery backup (unused)

GTP - float (UG482 TS-4)

Phmod 3A (bank 16 x5)

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

V10 10.139V_T2_D00_C00_R_34

FPGA Connections

PROJECT:	zuwu_a7	REV:	9/2/2026 1:29:21 PM
AUTHOR		Sheet:	6

